Properties

1. $X \sim N(\mu, \sigma^2) \Rightarrow E(X) = \mu$
$Var(X) = \sigma^2$

2. $X \sim N(\mu, \sigma^2) \Rightarrow aX + b \sim N(a\mu + b, a^2\sigma^2)$

Def: The standard normal r.v. is a r.v. denoted by Z, that is:

$$Z \sim N(0,1)$$

$$E(Z) = 0, \ Var(Z) = 1$$

Properties:

if $Z \sim N(0,1)$, then the CDF is:

$$\boxed{\phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{u^2}{2}} \, du}$$

↳ NEVER compute this!

Remark!

The connection between Z and X:

$Z \sim N(0,1)$
$X \in N(\mu, \sigma^2)$ $\Rightarrow$ $\boxed{Z = \frac{X-\mu}{\sigma}}$

$\Rightarrow$ the CDF of $X \in N(\mu, \sigma^2)$ is:

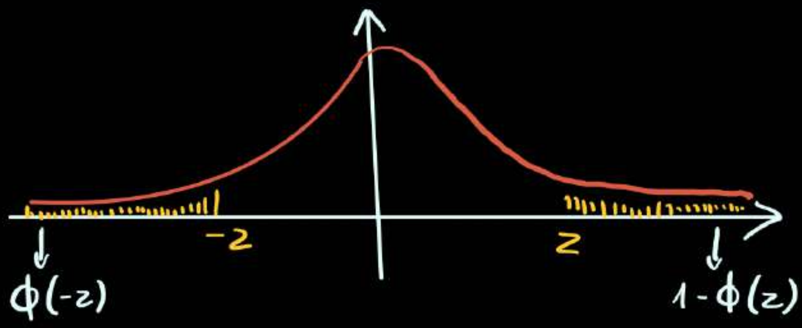
$$\boxed{F_X(x) = \phi\left(\frac{x-\mu}{\sigma}\right)}$$

Remark!

$$P(a < X \le b) = F_X(b) - F_X(a) = \phi\left(\frac{b-\mu}{\sigma}\right) - \phi\left(\frac{a-\mu}{\sigma}\right)$$

Properties

$$\phi(-z) = 1 - \phi(z)$$



Remark!

The tables for $\phi(z)$ are good for $|z| < 3$.

For regions with $|z| > 3$ we use the standard normal complementory CDF:

$$\boxed{Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty e^{-\frac{u^2}{2}} du = 1 - \phi(z)}$$

$a, b > 3\sigma_X \longrightarrow Q$

$a, b < 3\sigma_X \longrightarrow \phi$

Ex 1. $X \sim N(0,1)$, $Y \in N(0,2)$

Determine:

a. $P(-1 < X \le 1) = \phi(1) - \phi(-1) \overset{prop}{=} \phi(1) - (1 - \phi(1))$

$= 2\phi(1) - 1 = 0{,}683$

↳ we find the value in the table

b. $P(Y > 3{,}5)$

$3\sigma_Y = 3\sqrt{2} = 4{,}2 > 3{,}5 \Rightarrow$ use ϕ !!

$\Rightarrow P(Y > 3{,}5) = 1 - P(Y \le 3{,}5) = 1 - F_Y(3{,}5) = 1 - \phi\left(\frac{3{,}5 - 0}{\sqrt{2}}\right) =$

$= 1 - \phi(2{,}025) = 1 - 0{,}981 = 0{,}019$

c. $P(X > 3{,}5) = Q(3{,}5) = 2{,}33 \cdot 10^{-4}$

$3\sigma_x = 3 < 3{,}5 \Rightarrow$ use Q !!

d. $P(\underbrace{-1 < Y \le 1}_{< 3\sigma_y}) = F_Y(1) - F_Y(-1) = \phi\left(\frac{1-0}{\sqrt{2}}\right) - \phi\left(\frac{-1-0}{\sqrt{2}}\right)$

$= \phi\left(\frac{1}{\sqrt{2}}\right) - \underbrace{\phi\left(-\frac{1}{\sqrt{2}}\right)}_{= 1 - \phi\left(\frac{1}{\sqrt{2}}\right)} = 2\phi\left(\frac{1}{\sqrt{2}}\right) - 1 = 2\phi(0{,}7) - 1$

$\hookrightarrow$ use ϕ !!

3. Conditioning by an Event

Def: Let X - conditional r.v. with PDF $f_X(x)$

B - event, $P(B) > 0$

The Conditional PDF of X given B

$$\boxed{f_{X|B}(x) = \begin{cases} \dfrac{f_X(x)}{P(B)}, & x \in B \\ \text{zero, otherwise} \end{cases}}$$

Prop:

if $\{B_1, B_2, \dots B_n\}$ - event space, then:

$$\boxed{f_X(x) = P(B_1) f_{X|B_1}(x) + P(B_2) f_{X|B_2}(x) + \dots + P(B_n) f_{X|B_n}(x)}$$

Def: The conditional expected value of X given B:

$$E(X|B) = \int_{-\infty}^{\infty} x \cdot f_{X|B}(x)\, dx$$

Ex 2. PDF of X: $f_X(x) = \begin{cases} \frac{1}{10}, & x \in [0,10) \\ \text{zero, otherwise} \end{cases}$

Determine:

a. $P(X \le 6) = ?$ b. $f_{X|X\le 6}(x) = ?$ e. $E(X|X\le 6) = ?$

c. $P(X > 8) = ?$ d. $f_{X|X>8} = ?$ f. $E(X|X>8) = ?$

a. $P(X \le 6) = \int_{-\infty}^{6} f_X(x)\,dx = \int_{-\infty}^{0} 0\,dx + \int_0^6 \frac{1}{10}\,dx = \frac{x}{10}\Big|_0^6 = \frac{6}{10} = \boxed{\frac{3}{5}}$

b. $f_{X|X\le 6}(x) = \begin{cases} \dfrac{f_X(x)}{P(X\le 6)}, & x \le 6 \wedge x \in [0,10) \\ \text{zero, otherwise} \end{cases}$

$$\boxed{f_{X|X\le 6} = \begin{cases} \dfrac{\frac{1}{10}}{\frac{6}{10}} = \frac{1}{6}, & x \in [0,6] \\ \text{zero, otherwise} \end{cases}}$$

e. $E(X|X\le 6) = \int_{-\infty}^{0} x \cdot f_{X|X\le 6}(x)\,dx = \int_{-\infty}^{0} x \cdot 0\,dx + \int_0^6 x \cdot \frac{1}{6}\,dx + \int_6^{+\infty} x \cdot 0\,dx$

$= \frac{x^2}{2} \cdot \frac{1}{6}\Big|_0^6 = \frac{36}{2 \cdot 6} = \boxed{3}$

4. Probability Models of Derived R.V.

$X \sim$ a conditional r.v., $f_X(x) = $ PDF of X
$Y = g(X)$, g - continuous function

$\Rightarrow Y = g(X)$ - continuous r.v.

$f_X(y) = ?$ PDF of Y

- determine CDF of $X \to F_X(x)$
- determine CDF of $Y \to F_Y(y) = P(Y \le y)$
- determine $f_Y(y) = F_Y'(y)$

Ex ③ $X \sim U[-1,3]$, $Y = X^2$ (uniformly)

Determine $f_Y(y) = ?$

$$X \sim U[-1,3] \Rightarrow f_X(x) = \begin{cases} \frac{1}{4}, & x \in [-1,3] \\ \text{zero, otherwise} \end{cases}$$ PDF

$$F_X(x) = \int_{-\infty}^{x} f_X(x)\,dx$$

- $x < -1 \Rightarrow F_X(x) = \int_{-\infty}^{x} 0\,dx = 0$
- $x \in [-1,3] \Rightarrow F_X(x) = \int_{-\infty}^{-1} 0\,dx + \int_{-1}^{x} \frac{1}{4}\,dx = \frac{1}{4}(x+1)$
- $x > 3 \Rightarrow F_X(x) = \int_{-\infty}^{-1} 0\,dx + \int_{-1}^{3} \frac{1}{4}\,dx + \int_{3}^{x} 0\,dx = \frac{1}{4}x \Big|_{-1}^{3} = 1$

$$\Rightarrow F_X(x) = \begin{cases} 0, & x < -1 \\ \frac{x+1}{4}, & x \in [-1,3] \\ 1, & x > 3 \end{cases}$$

CDF of X

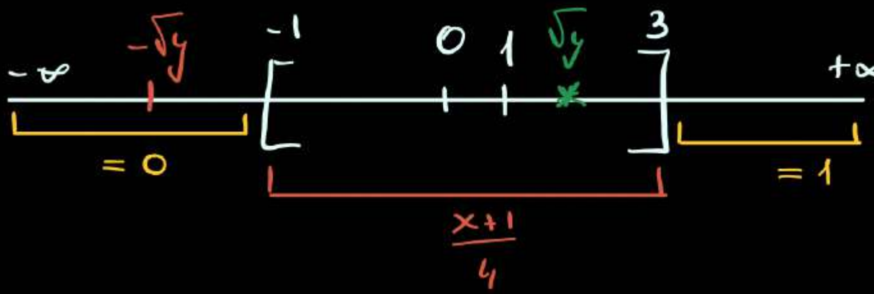
$$F_Y(y) = P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y}) =$$
$$= F_X(\sqrt{y}) - F_X(-\sqrt{y}) + \underbrace{P(X = -\sqrt{y})}_{=0}$$

I

$\sqrt{y} > 3$ ⟨ $-\sqrt{y} < -1$, $y > 9$

$$\Rightarrow F_Y(y) = \underbrace{F_X(\sqrt{y})}_{=1} - \underbrace{F_X(-\sqrt{y})}_{=0} = 1 - 0 = \boxed{1}$$

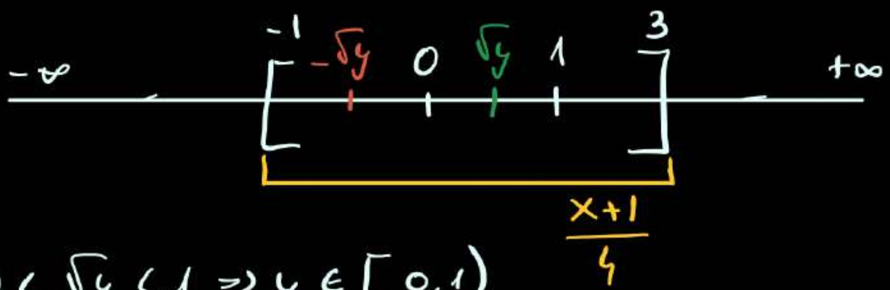
II



$\underline{1 < \sqrt{y} < 3} \iff y \in [1,9]$

$$F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y}) = \frac{\sqrt{y}+1}{4} - 0 = \boxed{\frac{\sqrt{y}+1}{4}}$$

III.



$\underline{0 < \sqrt{y} < 1} \Rightarrow y \in [0,1)$

$\Updownarrow$

$\underline{-\sqrt{y} > -1}$

$$F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y}) = \frac{\sqrt{y}+1}{4} - \frac{-\sqrt{y}+1}{4} = \frac{2\sqrt{y}}{4} = \boxed{\frac{\sqrt{y}}{2}}$$

II. $y<0 \Rightarrow F_Y(y) = 0$

$$F_Y(y) = \begin{cases} 1 , & y>9 \\ \frac{\sqrt{y}+1}{4} , & y \in [1,9] \\ \frac{\sqrt{y}}{2} , & y \in [0,1) \\ 0 , & y<0 \end{cases}$$

CDF of Y $\Rightarrow$

$$\Rightarrow f_Y(y) = F_Y'(y) = \begin{cases} \frac{1}{8\sqrt{y}} , & y \in [1,9] \\ \frac{1}{4\sqrt{y}} , & y \in (0,1) \\ 0, & \text{otherwise} \end{cases}$$

PDF of Y

Homework:

1. CDF of X: $F_X(x) = \begin{cases} a, & x \le -1 \\ \frac{x+1}{2} , & x \in (-1,1] \\ b , & x > 1 \end{cases}$

a. $a,b = ?$
b. PDF $= ?$
c. PDF of $Y = X^2$

2. PDF of X: $f_X(x) = \begin{cases} \frac{1}{5}, & x \in (-1,2) \\ \text{zero, otherwise} \end{cases}$

a. CDF $= ?$
b. PDF for $Y = e^X$
$Z = \sqrt{X}$